

Figure 3.3 D and L Alanine

3.2.5 Acid – base properties of amino acids

Amino acids have both carboxylic acid and amino groups. Similarly, the side chains of acidic and basic amino acids contain additional ionisable groups. Recall that acids are proton donors, where as bases are proton acceptors and the quantitative relationship between pH of a solution and the concentration of the weak acid and its conjugate base is given by Henderson-Hasselbalch equation.

The dissociation of the acidic and basic groups can be explained with alanine as an example. The three different ionic forms of alanine in acidic, neutral and basic solutions is explained in Figure 3.4

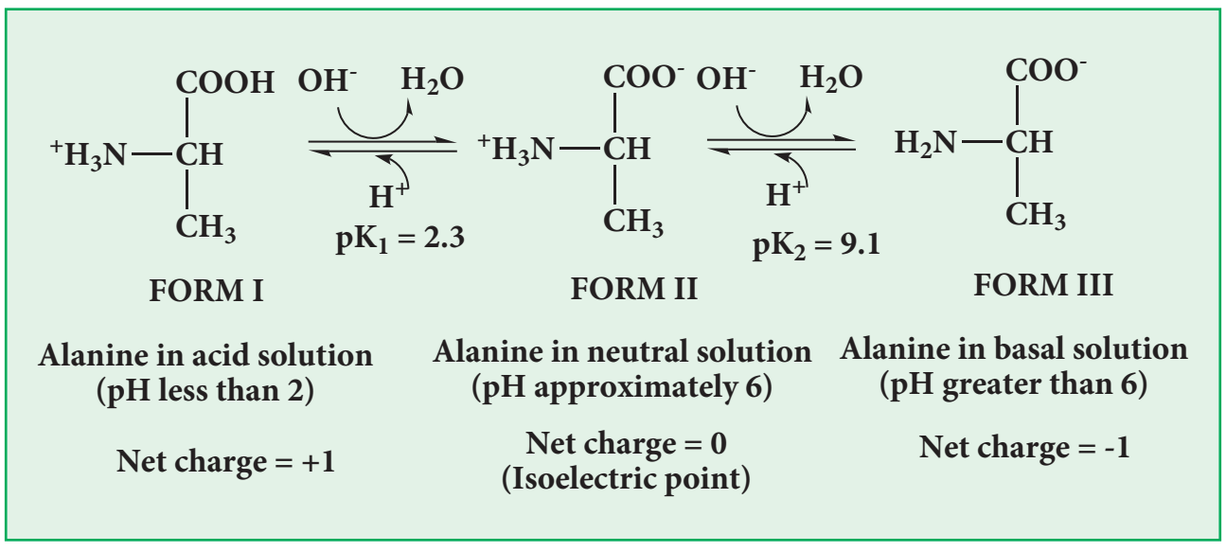


Figure 3.4 ionic forms of alanine

Note that at acidic pH both the amino and carboxylic acid groups are protonated (Form I) and as the pH increases the –COOH group dissociates to become Form II. At basic pH the $-NH_3^+$ group can dissociate to become Form III.

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